

JRAHS 2005 TRIAL HSC – MATHEMATICS (2 Unit)

Question 1.	[Start a New Page]	Marks
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(a)	Evaluate $\frac{8\pi}{2 + \sqrt[3]{2}}$ as a decimal correct to 3 significant figures.	1
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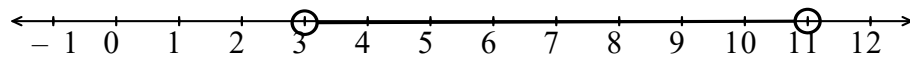
(b)	Find $\frac{d}{dx}[5x + \tan x]$.	2
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(c)	A group of cards are labelled 0, 1, 2, ... and 20. Find the probability that when a card is chosen, it is a prime number?	2
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(d)	Find $\int \sec 3x \tan 3x \, dx$, using the table of standard integrals.	2
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(e)	Find the exact value for $\sec \frac{\pi}{6}$.	1
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(f)	The number line graph represents the solutions to the inequality equation $ x - a \leq b$.	2
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Find the values of a and b .

(g)	Find integer p so that $(3 - \sqrt{5})^2 = 14 - \sqrt{p}$.	2
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Question 2. **[Start a New Page]**

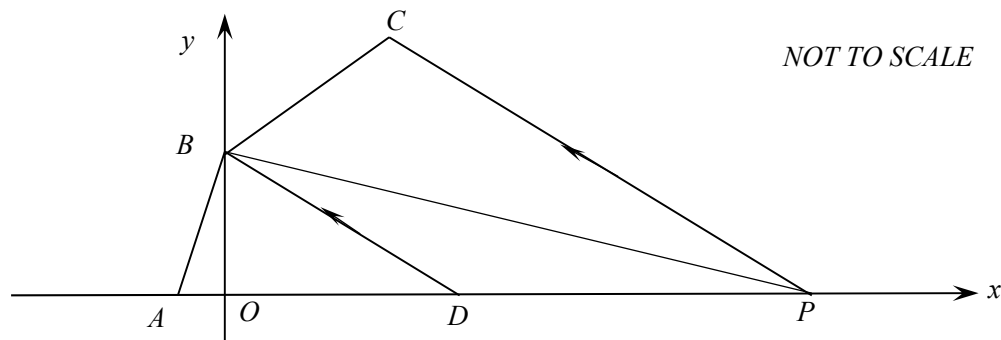
- (a) Differentiate with respect to x the following:
- (i) $\frac{\sin x}{x+1}$. **2**
- (ii) $\sqrt{1+e^{6x}}$. **2**
- (iii) $x^3 \ln x$. **2**
- (b) Find the domain for the function: $y = x + \ln(3-x)$. **1**
- (c) Solve for θ : $\tan \theta = 0.3$ (correct to two decimal places) for $0 < \theta < 2\pi$. **2**
- (d) Find a primitive function for $\frac{1}{3x}$. **1**
- (e) Evaluate $\int_0^2 (e^{-x} + 1) dx$, to two decimal places. **2**

Question 3.**[Start a New Page]****Marks**

- (a) Find the equation of the normal to the curve $y = 2\cos x + 3$ at the point $(\frac{\pi}{2}, 3)$.

2

- (b) The diagram shows the coordinates of four points:
 $A(-1, 0)$, $B(0, 2)$, $C(3, 5)$ and $D(8, 0)$.



- (i) Find the gradient of BD . **1**
- (ii) Find the equation of the line passing through B and D . **1**
- (iii) Find the angle of inclination of the line BD (to nearest the degree). **1**
- (iv) Given $CP \parallel BD$, show that the equation of CP is $x + 4y - 23 = 0$. **2**
- (v) Find the coordinates of point P , where the line CP intersects the x -axis. **1**
- (vi) Find the perpendicular distance of point C from the line segment BD . **2**
- (vii) Explain why the area of quadrilateral $ABCD$ is the same as the area of triangle ABP . **2**

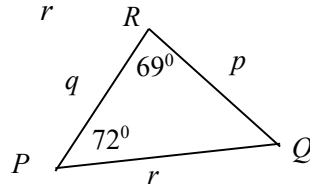
Question 4.

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Marks

- (a) For $\triangle PQR$, find $\frac{p}{r}$ correct to 3 decimal places.

2

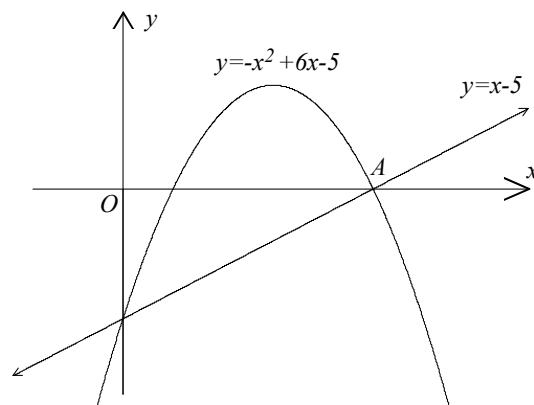


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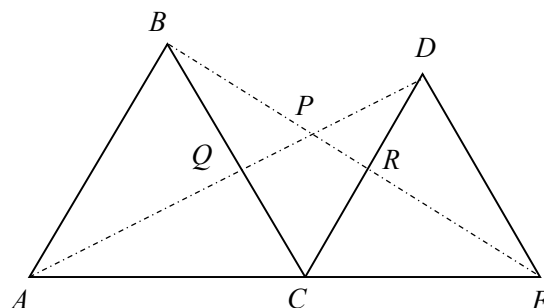
- (b) The roots of the equation $x^2 - 2x - 5 = 0$, are $x = \alpha$ and $x = \beta$.
Find the value of $\frac{1}{\alpha} + \frac{1}{\beta}$.

2

- (c) The diagram shows the sketch of the parabola $y = 6x - x^2 - 5$ and a line $y = x - 5$.



- (i) Find the x -coordinate of A . **1**
- (ii) Find the area of the shaded region bounded by the line and the parabola. **2**
- (d) Given the triangles ABC and CDE are different equilateral triangles.
 BE intersects AD at P , and A, C and E lie on the line segment AE .



NOT TO SCALE

- (i) Copy the diagram onto your writing booklet, and prove that $\triangle ACD \equiv \triangle ECB$.
- (ii) Show that $\angle APB = 60^\circ$.

3

2

Question 5.

[Start a New Page]

Marks

(a) Simplify: $1 + \cos^2 x + \cos^4 x + \dots$ for $0 < x < \frac{\pi}{2}$. **1**

(b) The first term of an Arithmetic series is a , the common difference is d and the n^{th} term is L .

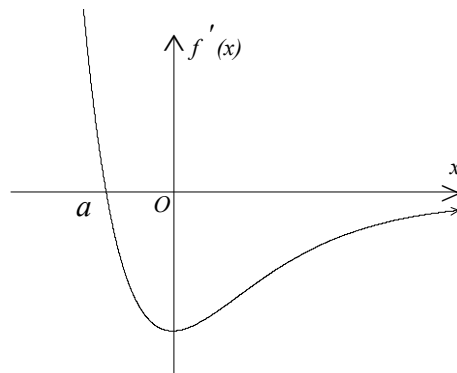
(i) Write down L , in terms of a , d and n . **1**

(ii) Show that the sum, S_n , of the first n terms can be expressed as **2**

$$S_n = \frac{(L+a)}{2} \left[1 + \frac{L-a}{d} \right].$$

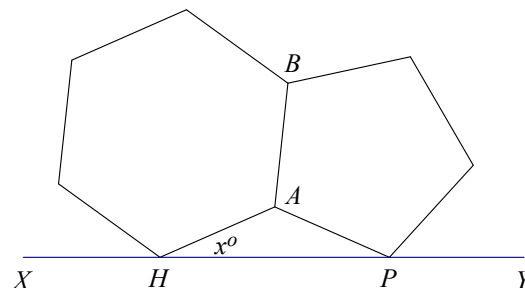
(iii) Hence, or otherwise, find: $5 + 8 + 11 + \dots + 173$. **1**

(c) The sketch of the Gradient function $f'(x)$ is shown below. **3**



Sketch the graph of the function, $y = f(x)$, given $f(x) > 2$ for $x > 0$.

(d) The figure consists of a regular hexagon and a regular pentagon, with a common side AB . **4**



Given vertex H and P lie on a straight line $XHPY$ and $\angle PHA = x^\circ$.

Copy the diagram onto your writing booklet and find the value of x . Give reasons.

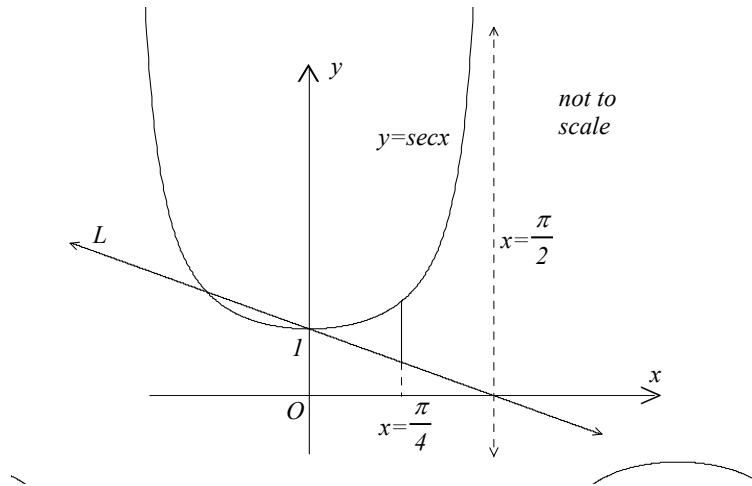
Question 6.	[Start a New Page]	Marks
(a)	Solve for x : $\log_3(2x - 5) = 1$.	2
(b)	Differentiate $f(x) = \frac{3}{x}$ with respect to x , by first principles.	2
(c)	A particle is moving in a straight line with velocity $v = 3e^t + 6e^{-t}$. It begins its motion at the Origin O , t is in seconds and v is in metres per second.	
(i)	What is its initial velocity?	1
(ii)	Is the particle ever at rest? Give reasons	1
(iii)	Find the displacement function, x , of the particle, at time t minutes.	2
(iv)	Find the time when the particle is at $x = 10$.	2
(d)	Sketch the graph of the curve $y = 3 \sin 2x$ for the interval $0 \leq x \leq \pi$.	2

Question 7.

[Start a New Page]

Marks

- (a) Given the sketch of the curve $y = \sec x$, for $0 \leq x < \frac{\pi}{2}$ and the line L as shown.



- (i) Show that the equation of the line, L , is $y = 1 - \frac{2}{\pi}x$. **1**
- (ii) The area enclosed by the curve $y = \sec x$, the lines L and $x = \frac{\pi}{4}$, is rotated about the x -axis. Find the exact volume for the solid of revolution. **3**
- (b) Use Simpson's rule, with three function values, to evaluate : $\int_0^4 \frac{3dx}{1 + \sqrt{x}}$, **2**
(correct to two decimal places).
- (c) Consider the function: $f(x) = x^3 - 3kx + 4$.
- (i) Explain why $f(x)$ is an increasing function for all x when $k < 0$. **2**
- (ii) Find the expression for each stationary point of $f(x)$, in terms of k , when $k > 0$. **2**
- (iii) Prove that $f(x)$ has 3 distinct real x -intercepts when $k^3 > 4$. **2**

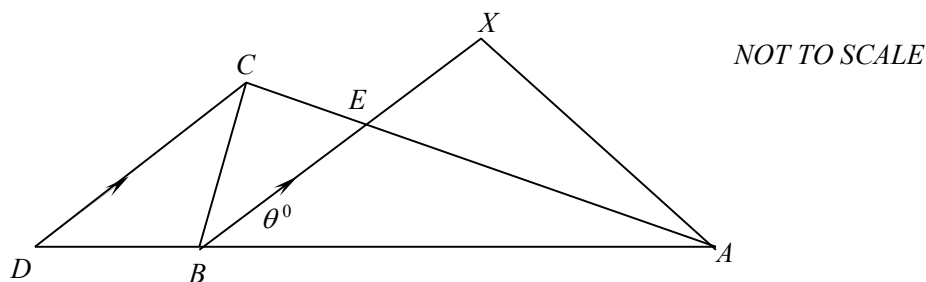
Question 8.	[Start a New Page]	Marks
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|-----|-----|--|----------|
| (a) | (i) | Show that the locus of all points $P(x, y)$, which are equidistant from the origin O and to the line $y = 4$, is the parabola: | 2 |
|-----|-----|--|----------|

$$x^2 = 16 - 8y.$$

- | | | | |
|--|------|---|----------|
| | (ii) | Hence, find the coordinates of the vertex V . | 1 |
|--|------|---|----------|

- | | |
|-----|---|
| (b) | In the diagram $BX \parallel DC$, XB bisects angle ABC and $AX \perp BX$ at X .
Let $\angle ABX = \theta^\circ$. |
|-----|---|



Copy this diagram onto your writing booklet.

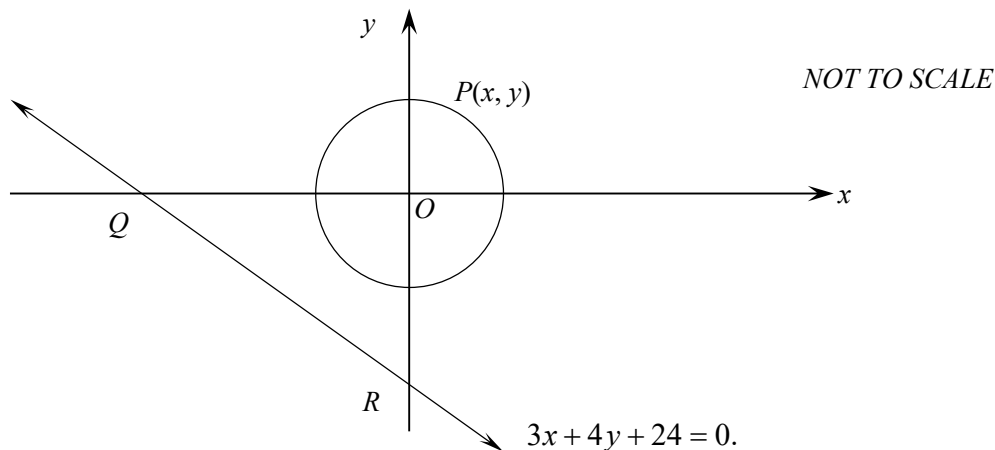
- | | | | |
|-------|--|--|----------|
| (i) | | State why $\angle BCD = \angle XBC$. | 1 |
| (ii) | | Show that triangle BCD is isosceles. | 2 |
| (iii) | | Hence, explain why $\frac{AE}{EC} = \frac{AB}{BC}$. | 2 |
| (iv) | | If $\frac{BA}{BC} = 3$, and by using the cosine rule, or otherwise, show that E is the midpoint of BX . | 4 |

Question 9.

[Start a New Page]

Marks

- (a) The mass of a substance X is M grams, at time t years.
It decays at an instantaneous rate proportional to its mass at time t years,
ie $\frac{dM}{dt} = -kM$, where k is the decay rate constant of proportionality.
- (i) Verify that $M = M_0 e^{-kt}$ satisfies the rate equation $\frac{dM}{dt} = -kM$,
where M_0 is the initial mass of substance X . **1**
- (ii) Hence, show that the time, T years, for half the mass of X to decay,
is given by $T = \frac{\ln 2}{k}$. **1**
- (iii) Find the decay rate constant, if the half-life of substance X is 3 466 years? **1**
- (iv) Sketch the graph of $\frac{dM}{dt}$ against M . **1**
- (b) Consider the circle: $x^2 + y^2 = 1$ and the line: $3x + 4y + 24 = 0$.
The points Q and R are the x and y -intercepts of the line $3x + 4y + 24 = 0$.
The point $P(x, y)$ lies on the circle as shown in the diagram.



- (i) As P moves around the circle, show that the perpendicular distance, W ,
from the length QR to the circle, is given by: $W = 3x + 4y + 24$. **3**
- (ii) Hence, or otherwise, find the least length of W . **5**

Question 10. **[Start a New Page]** **Marks**

- (a) Mary visits the sock section of a shop that has 5 different pairs of socks individually arranged on a table.
She randomly selects socks one at a time.
- (i) Explain why the probability that Mary does *not* have a matching pair of socks, after selecting the second sock, is $\frac{8}{9}$. **1**
- (ii) Find the probability she does *not* have a matching pair of socks after selecting the third sock. **2**
- (iii) What is the probability that, in the first 3 socks, Mary does have a matching pair? **1**
- (b) Mr Howzat borrows \$30 000 from a bank. Interest is to be calculated at 12% *pa*, compounded monthly, on the balance remaining over the term of the loan of 7 years.
Each year, at k regular intervals, (where $k = 1, 2, 3, \dots$ or 12), Mr Howzat repays \$ F for each instalment.
- (i) Show that the amount owing, \$ A_2 , after the second instalment is paid, is given by: **2**
- $$A_2 = 30\,000 \left[(1.01)^{\frac{12}{k}} \right]^2 - F \left[1 + (1.01)^{\frac{12}{k}} \right].$$
- (ii) Show that the amount of each instalment, \$ F , is given by: **3**
- $$F = 30\,000 \times 1.01^{84} \times \frac{[1.01^{\frac{12}{k}} - 1]}{[1.01^{84} - 1]}.$$
- (iii) Calculate the value of each instalment if the instalments are made quarterly ($k = 4$). **1**
- (iv) How much would Mr Howzat have saved over the term of the loan if he had chosen to make monthly rather than quarterly instalments? **2**

THE END



Q1(a) $\frac{8\pi}{2+3\sqrt{2}} = 7.7046...$ $(2+3\sqrt{2}) = 7.71$ (3 sf) ①	1	Q2(a) (i) $y = \frac{\sin x}{x+1}$ ② $\therefore \frac{dy}{dx} = \frac{\cos x \cdot (x+1) - 1 \cdot \sin x}{(x+1)^2}$ $= \frac{(x+1)\cos x - \sin x}{(x+1)^2}$	1, 1
(b) $\frac{d}{dx} [5x + \tan x]$ $= 5 + \sec^2 x$ ②	1, 1	(ii) $y = \sqrt{1+e^{6x}} = (1+e^{6x})^{\frac{1}{2}}$ $\frac{dy}{dx} = \frac{1}{2} (1+e^{6x})^{-\frac{1}{2}} \cdot 6e^{6x}$ $= \frac{3e^{6x}}{\sqrt{1+e^{6x}}}$ ②	1, 1
(c) $P(\text{prime}) = \frac{8}{21}$ ② $\{2, 3, 5, 7, 11, 13, 17, 19\}$	1, 1	(iii) $y = x^3 \ln x$ ② $\frac{dy}{dx} = 3x^2 \ln x + x^3 \cdot \frac{1}{x}$ $= 3x^2 \ln x + x^2$	1, 1
(d) $\int \sec 3x \tan x \, dx$ ② $= \frac{1}{3} \sec 3x + C$	1, 1	(iv) For $0^\circ < x < 90^\circ$ ① $\therefore \text{O.f.} = \{x : x < 3\}$	1
(e) $\sec \frac{\pi}{6} = \sec 30^\circ$ ① $= \frac{2}{\sqrt{3}}$	1	(v) $\tan \theta = 0.3$ ② $\theta = 0.2914... \text{ or } \pi + 0.2914...$ $\therefore \theta = 0.29 \text{ or } 3.43$ (2 dp) ②	1, 1
(f) $a = \frac{3+4}{5} = \frac{7}{5}$ ② $b = 7-3 = 4$	1	(d) $\int \frac{1}{3x} \, dx = \frac{1}{3} \ln x + C$ $= \frac{1}{3} \ln(3x) + C$	1
(g) $(3-\sqrt{5})^2 = 9 - 6\sqrt{5} + 5$ $= 14 - 6\sqrt{5}$ $\therefore 6\sqrt{5} = \sqrt{p}$ $p = 36 \times 5 = 180$ ②	1	(e) $\int_0^2 e^{-x} + 1 \, dx$ $= [-e^{-x} + x]_0^2$ $= (-e^{-2} + 2) - (-e^0 + 0)$ $= -e^{-2} + 2 + 1$ $= 3 - e^{-2}$	1

Q3(a) $y = 2 \cos x + 3$
 $\frac{dy}{dx} = -2 \sin x$
at $x = \frac{\pi}{3}$ $m_T = -2 \sin \frac{\pi}{3} = -2$
 $\therefore m_N = \frac{1}{2}$
Equ. of normal $(\frac{\pi}{3}, 3)$
 $y - 3 = \frac{1}{2} (x - \frac{\pi}{3})$
 $y = \frac{1}{2}x + 3 - \frac{\pi}{4}$ / $2x - 4y + 12 - \pi = 0$ ②

(b) (i) $m_{OD} = \frac{2-0}{0-8} = -\frac{1}{4}$ ①

(ii) Equ. of OD: $y = -\frac{1}{4}x + 2$
 $B(0, 2)$ $[x + 4y - 8 = 0]$ ①

(iii) $m = \tan \theta = -\frac{1}{4}$
 $\therefore \theta = 165^\circ 58'$
 $= 166^\circ$ ①

(iv) $m_{CP} = m_{BD} = -\frac{1}{4}$ ②
Equ. of CP: $y - 5 = -\frac{1}{4}(x - 3)$
 $4y - 20 = -x + 3$
 $\therefore x + 4y - 23 = 0$

(v) at P $y = 0$
 $\therefore x + 0 - 23 = 0 \Rightarrow x = 23$
 $\therefore P = (23, 0)$

(vi) $\perp$ dist. $= \frac{|1 \times 3 + 4 \times 5 - 8|}{\sqrt{1^2 + 4^2}}$
 $C = (3, 5)$
 $x + 4y - 8 = 0$
 $= \frac{15}{\sqrt{17}}$ units ②

(ii) Now $|ABCD| = |ABD| + |BDC|$
but $|BDP| = |BDC|$ (same base BD, same perp. height between parallel lines)
 $\therefore |ABCD| = |ABD| + |BDP|$
 $= |ABP|$ ②

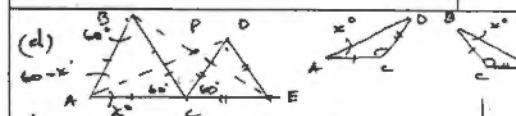
OR $|APB| = \frac{1}{2} \times 24 \times 2 = 24 \text{ u}^2$
 $|ABD| = \frac{1}{2} \times 9 \times 2 = 9 \text{ u}^2$ (common)

Q4(a) $\frac{p}{\sin 72^\circ} = \frac{r}{\sin 69^\circ}$
 $\therefore \frac{p}{r} = \frac{\sin 72^\circ}{\sin 69^\circ} = 1.018719436...$
 $= 1.019$ (3 dp) ②

(b) $x^2 - 2x - 5 = 0$ ②
 $\frac{1}{a} + \frac{1}{b} = \frac{a+b}{ab} = \frac{-2}{-5} = \frac{2}{5}$ ②

(c) (i) $y = x - 5$
at A $y = 0$ $\therefore x = 5$

(ii) Area $= \int_0^5 (y_U - y_L) \, dx$
 $= \int_0^5 (5x - x^2 - 5) \, dx$
 $= \int_0^5 (5x - x^2 - 5) \, dx$
 $= \left[\frac{5}{2}x^2 - \frac{1}{3}x^3 - 5x \right]_0^5$
 $= \frac{125}{2} - \frac{125}{3} - 25$
Area $= \frac{125}{6} = 20\frac{5}{6} \text{ u}^2$ ②

(d) 
(i) In Δ s ACD and ECD
1. $AC = EC$ (all sides equal in equilateral ΔABC)
2. $\angle ACD = \angle ECB = \angle BCD + 60^\circ$
 $\therefore$ (all angles are 60° in equilateral Δ)
3. $CD = CE$
 $\therefore \Delta ACD \cong \Delta ECD$ (SAS) ②

(ii) Let $\angle CAD = x^\circ$
1. $\angle CBE = x^\circ$ (corresp. angles in cong. triangles are equal)
2. $\angle BAP = 60^\circ - x^\circ$
3. $\angle APB + 60^\circ - x^\circ + 60^\circ + x^\circ = 180^\circ$ (angle sum of ΔAPB is 180°)
 $\therefore \angle APB = 60^\circ$ ②

* $BD = \sqrt{65}$
and $|BDC| = \frac{1}{2} \times \sqrt{65} \times \frac{15}{\sqrt{17}} = 15 \text{ u}^2$
 $|BDP| = \frac{1}{2} \times \sqrt{65} \times \frac{15}{\sqrt{17}} = 15$
 $\therefore |ABCD| = 9 + 15 = 24$
 $\therefore |ABCD| = |ABP| = 24 \text{ u}^2$

6

$$\log_3(2x-5) = 1$$

$$\therefore 2x-5 = 3^1 = 3$$

$$\therefore x = 4$$

$$f'(x) = \lim_{h \rightarrow 0} \left(\frac{\frac{3}{x+h} - \frac{3}{x}}{h} \right)$$

$$= \lim_{h \rightarrow 0} \left(\frac{3x - 3(x+h)}{hx(x+h)} \right)$$

$$= \lim_{h \rightarrow 0} \frac{-3}{x(x+h)} \quad \begin{matrix} h \neq 0 \\ \text{and} \\ h \rightarrow 0 \end{matrix}$$

$$= \lim_{h \rightarrow 0} \frac{-3}{x(x+h)}$$

$$f'(x) = \frac{-3}{x(x+h)} = \frac{-3}{x^2}$$

$$v = 3e^t + 6e^{-t} \quad \begin{matrix} t=0 \\ x=0 \end{matrix}$$

$$\text{initial velocity} = 3+6 = 9 \text{ m/s}$$

$$e^t, e^{-t} > 0 \text{ for all } t \geq 0$$

$$v = 3e^t + 6e^{-t} > 0$$

$$\text{particle never at rest, as } v \neq 0$$

$$x = \int (3e^t + 6e^{-t}) dt$$

$$x = 3e^t - 6e^{-t} + c$$

$$\text{at } t=0, x=0$$

$$\therefore 0 = 3 - 6 + c$$

$$\therefore c = 3$$

$$\text{so } x = 3e^t - 6e^{-t} + 3$$

$$\therefore x = 3e^t - 6e^{-t} + 3 = 10$$

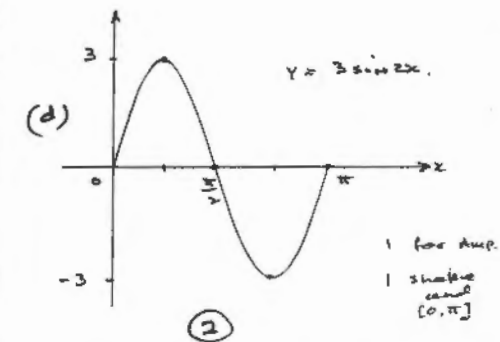
$$3e^t - 6e^{-t} - 7 = 0$$

$$3e^{2t} - 6 - 7e^t = 0$$

$$u = e^t$$

$$3u^2 - 7u - 6 = 0$$

$$(3u+2)(u-3) = 0$$



$$\therefore u = -\frac{2}{3} \text{ or } u = 3$$

$$\text{i.e. } e^t = -\frac{2}{3} \text{ or } e^t = 3$$

$$\text{but } e^t > 0 \text{ or } t = \ln 3$$

$$\therefore \text{no solution possible}$$

$$\therefore \text{the time to be at } x=10$$

$$\text{is } \ln 3 \text{ seconds}$$

4

Q5.

$$r = \cos^2 x \quad 0 \leq x < \frac{\pi}{2}$$

$$\therefore s = \frac{a}{1-r} = \frac{1}{1-\cos^2 x}$$

$$s = \frac{1}{\sin^2 x} = \sec^2 x$$

$$(b) (i) T_n = L = a + (n-1)d$$

$$(ii) S_n = \frac{n}{2} [2a + (n-1)d] \parallel \frac{n}{2} [a+L]$$

$$\text{but as } L = a + (n-1)d$$

$$(n-1) = \frac{L-a}{d}$$

$$n = \frac{L-a}{d} + 1$$

$$\text{so } S_n = \frac{1}{2} \left(\frac{L-a}{d} + 1 \right) [2a + \left(\frac{L-a}{d} \right) d]$$

$$= \frac{1}{2} \left(1 + \frac{L-a}{d} \right) [2a + L - a]$$

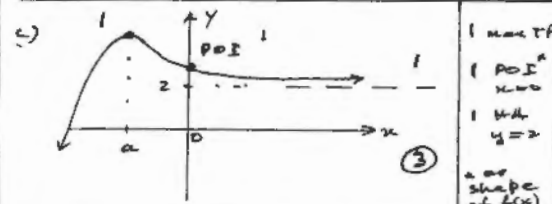
$$\therefore S_n = \frac{1}{2} (a+L) \left(1 + \frac{L-a}{d} \right)$$

$$(iii) \text{Hence:}$$

$$S = \frac{1}{2} (5+173) \left(1 + \frac{173-5}{3} \right)$$

$$= \frac{1}{2} \times 178 \times (1+56)$$

$$\text{Sum} = 5073$$



$$(d) 1. \angle BAH = \left(\frac{6-2}{2} \right) \times 120^\circ = 120^\circ$$

$$(\text{all angles of reg. hexagon } 120^\circ)$$

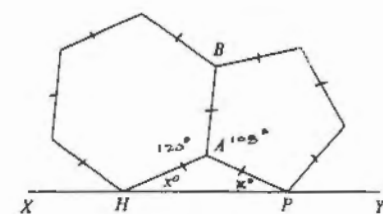
$$2. \angle BAP = \left(\frac{5-2}{2} \right) \times 120^\circ = 108^\circ$$

$$(\text{all angles of reg. pentagon } 108^\circ)$$

$$3. \angle HAP + 120^\circ + 108^\circ = 360^\circ$$

$$(\text{angle sum at pt. A is } 360^\circ)$$

$$\therefore \angle HAP = 132^\circ$$



$$4. AH = AP = AB \quad (\text{all sides of reg. hexagon + pentagon are equal})$$

$$5. \angle APH = \angle AHP = x^\circ \quad (\text{equal angles opposite equal sides})$$

$$6. x + x + 132 = 180 \quad (\text{angle sum of } \triangle AHP = 180^\circ)$$

$$2x = 48$$

$$x = 24$$

3

a) (i) $OP = PM$



$$\sqrt{x^2 + y^2} = |4 - y|$$

$$x^2 + y^2 = (4 - y)^2$$

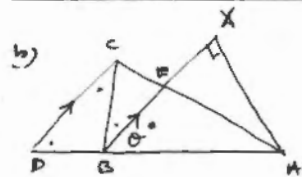
$$x^2 + y^2 = 16 - 8y + y^2$$

$$x^2 = 16 - 8y$$

②

ii) $x^2 = 16 - 8y$
 $x^2 = 4(-2)(y - 2)$

$\therefore V = (0, 2)$ ①



(i) $\angle BCD = \angle XBC$
 Alternates angles equal
 as $DC \parallel BX$ ①

ii) 1. $\angle CBX = 90^\circ$ (Data)
 2. $\angle CDB = 90^\circ$ (Corresponding angles equal as $DC \parallel BX$)
 $\therefore \angle CDB = \angle DCB = 90^\circ$
 $\therefore \triangle BCD$ is isosceles (two angles equal) ②

iii) $\frac{AE}{EC} = \frac{AB}{BD}$

but $DB = CB$ (equal sides opposite equal angles)

$\therefore \frac{AE}{EC} = \frac{AB}{CB}$ Q.E.D.

②

(i) $m_L = \frac{1-0}{0-\frac{2}{\pi}} = -\frac{\pi}{2}$

(0, 1)
 $(\frac{2}{\pi}, 0)$

$\therefore$ Eqn. of L: $y = -\frac{\pi}{2}x + 1$.

(ii) Volume = $\pi \int_0^{\pi/4} \sec^2 x dx - \pi \int_0^{\pi/4} (1 - \frac{\pi}{2}x)^2 dx$

$$= \pi \int_0^{\pi/4} \sec^2 x - (1 - \frac{\pi}{2}x)^2 dx$$

$$= \pi \int_0^{\pi/4} \sec^2 x - 1 + \frac{\pi}{2}x - \frac{\pi^2}{4}x^2 dx$$

$$= \pi \left[\tan x + \frac{\pi}{6} (1 - \frac{\pi}{2}x)^2 \right]_0^{\pi/4}$$

$$= \pi \left[\tan x - x + \frac{\pi}{2}x^2 - \frac{\pi^2}{4}x^3 \right]_0^{\pi/4}$$

$$= \pi \left[(1 + \frac{\pi}{2} (1 - \frac{\pi}{2})^2) - (0 + \frac{\pi}{2}) \right]$$

$$= \pi \left[1 + \frac{\pi}{2} - \frac{\pi^2}{4} \right]$$

③

Vol = $\pi \left[1 - \frac{1\pi}{48} \right] \pi^3$

(b) $I = \int_0^4 \frac{3x}{1+\sqrt{x}}$

x	0	2	4
$\frac{3}{1+\sqrt{x}}$	3	$\frac{3}{1+\sqrt{2}}$	1

$\therefore I = \frac{2}{3} \left[3 + 1 + 4 \times \frac{3}{1+\sqrt{2}} \right]$

$$= \frac{2}{3} \times 8.91056...$$

$$= 5.940375...$$

②

$\therefore I = 5.94$ (2 dp)

(c) (i) $f(x) = 3x^2 - 3k$

If $k < 0$ then $3x^2 - 3k > 0$, $3x^2 > 0$

so $f(x) = 3x^2 - 3k > 0$

$\therefore f(x)$ is increasing $\forall x$.

②

(c) (ii)

For S.P.s to occur $f'(x) = 0$

$\therefore 3x^2 - 3k = 0$

$x^2 = k$

$k > 0$ $x = \sqrt{k}$ or $-\sqrt{k}$

so $f(\sqrt{k}) = (\sqrt{k})^3 - 3k\sqrt{k} + 4$
 $= k\sqrt{k} - 3k\sqrt{k} + 4$
 $= 4 - 2k\sqrt{k}$

and $f(-\sqrt{k}) = (-\sqrt{k})^3 - 3k(-\sqrt{k}) + 4$
 $= -k\sqrt{k} + 3k\sqrt{k} + 4$
 $= 4 + 2k\sqrt{k}$

$\therefore$ S.P.s are $(\sqrt{k}, 4 - 2k\sqrt{k})$
 ② $(-\sqrt{k}, 4 + 2k\sqrt{k})$

(iii) Since a cubic
 For 3 distinct real roots

$\Delta > 0$ $y_1, y_2 < 0$

so $(4 - 2k\sqrt{k})(4 + 2k\sqrt{k}) < 0$
 $16 - 4k^2 < 0$

$16 - 4k^2 < 0$
 $-4k^2 < -16$
 $\therefore k^2 > 4$

② as $k > 0$
 $4 + 2k\sqrt{k} > 0$

but $4 - 2k\sqrt{k} < 0$ for 3 roots
 $\therefore 2k\sqrt{k} > 4$

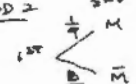
$k\sqrt{k} > 2$
 $\therefore k^3 > 4$

210.

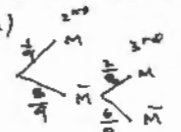
) (i) METHOD 1, Not match.

$$P(E) = 10 \times \frac{1}{10} \times \frac{10-2}{10-1} = \frac{8}{9}$$

METHOD 2



$$P(E) = \frac{8}{9} \quad (1)$$

i) 

$$P(E = \text{no match after 3 socks}) = \frac{8}{9} \times \frac{6}{8} = \frac{2}{3} \quad (2)$$

i) $P(E = \text{a matching pair in 3}) = 1 - P(\text{no match}) = 1 - \frac{2}{3} = \frac{1}{3} \quad (1)$

$$= P(MM) + P(M\bar{M}) + P(\bar{M}M) = \frac{1}{3}$$

) (i) $n = 10$ p withinterval k gives $\frac{12}{k}$ $\therefore n = 7k$

Amount owing at end of 1st pf

$$R_1 = 30000 \times 1.01^{12/k} - F$$

Amount owing at end of 2nd pf

$$R_2 = R_1 \times 1.01^{12/k} - F$$

$$= 30000 \times 1.01^{2 \times 12/k} - F(1 + 1.01^{12/k})$$

$$R_n = 30000 \times [1.01^{12/k}]^n - F[1 + 1.01^{12/k}]$$

Amount owing at end of n^{th} pf

$$R_n = 30000(1.01^{12/k})^n - F[1 + 1.01^{12/k} + \dots + 1.01^{(n-1)12/k}]$$

but when $\frac{12}{k} \times n = 84$ or $n = 7k$

$$R_{7k} = 0 \text{ debt paid}$$

Q 10 (b) (ii)

$$0 = 30000 \times 1.01^{84} - F[1 + 1.01^{12/k} + \dots + 1.01^{(7k-1)12/k}]$$

$$0 = 30000 \times 1.01^{84} - F \times \frac{(1.01^{12/k})^n - 1}{1.01^{12/k} - 1}, \quad n = 7k$$

$$\therefore F \times \left[\frac{(1.01^{12/k})^n - 1}{1.01^{12/k} - 1} \right] = 30000 \times 1.01^{84}$$

$$\text{i.e. } F \times \left[\frac{1.01^{84} - 1}{1.01^{12/k} - 1} \right] = 30000 \times 1.01^{84}$$

$$\therefore F = \frac{30000 \times 1.01^{84} (1.01^{12/k} - 1)}{(1.01^{84} - 1)}$$

$$(iii) \quad k = 4, \quad \frac{12}{k} = 3$$

$$\therefore 3 \text{ months}$$

$$F = 1604.69$$

$$(iv) \quad k = 12, \quad \frac{12}{k} = 1$$

$$F = 529.58$$

$$\text{Save} = 2BF_4 - 84F_{12}$$

$$= 446.33$$

Q9.

(a) (i) $\frac{dM}{dt} = -KM$

LHS RHS

$$\frac{d}{dt} M_0 e^{-Kt} = -KM_0 e^{-Kt}$$

$$\therefore LHS = RHS$$

$$\therefore M = M_0 e^{-Kt} \text{ satisfies } \frac{dM}{dt} = -KM$$

(ii) $t = T, \quad M = \frac{1}{2} M_0$

$$\therefore \frac{1}{2} M_0 = M_0 e^{-KT}$$

$$\frac{1}{2} = e^{-KT}$$

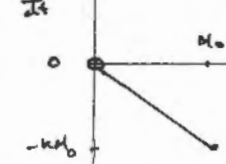
$$\ln \frac{1}{2} = -KT$$

$$T = \frac{\ln \frac{1}{2}}{-K} = -\frac{\ln 2}{-K} = \frac{\ln 2}{K}$$

(iii) $3466 = \frac{\ln 2}{K}$

$$K = \frac{\ln 2}{3466} = 2 \times 10^{-4} \text{ (1SF)}$$

(iv) $\frac{dM}{dt}$



$$\text{gradient} = -K$$

$$\text{negative gradient line segment}$$

b) (i) $W = \frac{3x + 4y + 24}{\sqrt{3^2 + 4^2}}$

$$= \frac{3x + 4y + 24}{5}$$

$$\therefore W = \frac{3x + 4y + 24}{5} \text{ reqd.}$$

$$(2)$$

Q9 (b) (ii) METHOD 1

$$P(x, y) \text{ lies on } x^2 + y^2 = 1$$

$$\therefore y = \pm \sqrt{1-x^2}$$

not shortest distance

will occur on lower semi-circle

$$\therefore y = -\sqrt{1-x^2}$$

$$\therefore W = \frac{1}{5} (3x - 4\sqrt{1-x^2} + 24)$$

$$\frac{dW}{dx} = \frac{1}{5} [3 - 4x \frac{1}{2} (1-x^2)^{-\frac{1}{2}} - 2x]$$

$$= \frac{1}{5} [3 + \frac{4x}{\sqrt{1-x^2}}]$$

$$\text{For possible max/min. values of } W \text{ to occur } \frac{dW}{dx} = 0$$

$$\therefore 3 + \frac{4x}{\sqrt{1-x^2}} = 0$$

$$\therefore \frac{4x}{\sqrt{1-x^2}} = -3 \sqrt{1-x^2}$$

$$10 \times 10 \text{ for } 20 \text{ units}$$

$$\therefore 16x^2 = 9(1-x^2)$$

$$16x^2 = 9 - 9x^2$$

$$25x^2 = 9$$

$$x^2 = \frac{9}{25}$$

$$x = \pm \frac{3}{5}$$

$$\text{but } x < 0 \quad x = -\frac{3}{5}, y = -\frac{4}{5}$$

$$\text{TEST at } x = -\frac{3}{5}$$

x	-0.7	$-\frac{3}{5}$	$-\frac{1}{5}$
W'	-0.18	0	0.138
	$\ominus$	$-$	$\oplus$

$$\text{Since } W \text{ is const + diffble over and } \frac{dW}{dx} \text{ changes sign } (- \rightarrow +)$$

$$\therefore \text{a relative min. T.P. at } x = -\frac{3}{5}$$

$$\text{Since no other T.P.s in } -1 \leq x \leq 1$$

$$\therefore \text{(abs) min. length is}$$

$$(5) \quad W = \frac{1}{5} [3(-\frac{3}{5}) + 4(-\frac{4}{5}) + 24]$$

$$= 3.8 \text{ units}$$